



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Trimester: Fall 2025 Marks: 40 Time: 2 Hours

Code: CSE 3411 Course Title: System Analysis & Design

“Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.”

Answer all the following questions:

Question 1:

[CO3]

[18]

Consider the following scenario:

Client (Metro Rail Authority): I want to build a system for each metro station. The overall purpose of the system is to control station-level operations and coordinate with a central control server. The system will be called the **Intelligent Station Control System (ISCS)** and will run independently at every station.

System Analyst: What will the station system control?

Client: The system will manage entry and exit gates, platform information displays, and train arrival handling. It must remain in a LOCKED state until a station officer logs in. Without authentication, no station operation is allowed.

System Analyst: How does the system behave during normal operation?

Client: After login, the system enters a READY state. Passengers can enter using smart cards or QR tickets. Validation is done locally and then synchronized with the Central Control Server. Before a train arrives, the server sends a prepare-to-arrive message. The system moves to PREPARE_ARRIVAL, updates display, and restricts platform access. When the train stops, the system enters BOARDING.

System Analyst: What about abnormal situations?

Client: If passenger density exceeds a limit, a gate malfunctions, or an emergency alert is triggered, the system immediately switches to CONTROLLED_SHUTDOWN. Passenger entry is blocked, warning messages are displayed, and a report is sent to the server.

System Analyst: How does the system resume operation?

Client: The server replies with one of three instructions: resume operation, partial restriction, or evacuate platform. The station system must act accordingly. After the train departs and all checks are completed, the system returns to READY.

System Analyst: Can administrators directly operate station devices?

Client: No. They can only monitor station status and receive reports during live operation.

A) Draw the Use Case Diagram of the Intelligent Station Control System (ISCS) from the station (client-side) perspective, showing at least one *include* and one *extend* relationship. [4]

B) Draw the Class Diagram and 2 major class **CRC** card of the ISCS station-side system. [4 + 2]

C) Draw the Sequence Diagram of the complete ISCS operation flow. [4]

D) Draw the State Diagram representing the different states of the ISCS starting from system lock to post-departure readiness. [4]

Question 2: [CO4] [5]

Mr. Akhshay Kumar, a small workshop owner who is not confident with money matters, is asked by a bank officer from “Laxmi Chit Fund” to invest in an online food ordering app for small restaurants in Dhaka, where many still take orders manually, claiming that it is a “*sure profit*”. Mr. Akhshay asks you (a financial expert) to verify the claim using the NPV method. The annual discount rate is 12%. The bank officer claims “*Sir, the project is definitely profitable even before year 3.*”

The project requires an initial investment of \$20,000 and has a duration of 5 years. The operating cost in Year 1 is \$4,000. The operating cost in Year 2 is 25% more than Year 1. The operating cost in Year 3 is 40% of the Year 2 cost, while the operating cost in Year 4 is equal to the average of Year 1 and Year 3 costs. The operating cost in Year 5 is twice the Year 3 cost. The project generates the same revenue amount **\$A** at the end of Year 1, Year 3, and Year 5, and no revenue is generated in Year 2 and Year 4.

- A) Using NPV method, determine the value of **A** for which the project is at break-even. [3]
- B) Using the result from 2(A), verify whether the officer’s claim is true or not. [1]
- C) Apart from economic feasibility, discuss **any other type of feasibility** Mr. Akshay should consider before investing in the online food ordering and delivery system. [1]

Question 3: [CO3] [12]

A) Select any **two** characteristics of a good SRS. For each, briefly explain the characteristics and show how it helps reduce errors, misunderstandings, or rework. [3]

B) Explain why involving stakeholders in SRS preparation is important and give two examples of problems that may arise if stakeholders are not consulted. List two functional and two non-functional requirements for the scenario in question 01. [2+2]

C) Create the UI designs for the project described in **Question 1**. Along with other features mentioned in the scenario, the UI should also show train schedules, gate status, passenger alerts, and allow staff to respond to emergencies. [3]

D) Explain how and which Rule/Concept of UI Design did you consider while designing the UI in C? [2]

Question 4: [CO3] [5]

Write down an SRS document and present it as a group having the following parts on your software project work covering: Introduction, System Study, System Analysis, Feature list fixation with Functional and Non-functional requirements, Different Feasibility Analysis, System design (UML diagrams), UI design through Figma software, implementation plan etc. (**no need to answer this question in the examination**, it has already been evaluated in classroom activities). [5]